

Lab Program 7: Java Method Overriding

Demonstrating Dynamic Method Dispatch & Inheritance with Bank Interest Rates

1. Objective

To demonstrate Object-Oriented Programming principles in Java—specifically **Inheritance**, **Method Overriding**, and **Dynamic Method Dispatch (Runtime Polymorphism)**. The program calculates interest rates for different banks (SBI, ICICI, PNB) overriding the base guidelines defined by the central bank (RBI).

2. Systems Requirements

Component	Requirement Details
Operating System	Windows 10/11, macOS, or Linux (Ubuntu/Debian)
Software / JDK	Java Development Kit (JDK) 8.0 or higher
IDE / Editor	Eclipse, IntelliJ IDEA, VS Code, or Command Line Terminal
Processor	Intel Core i3 / AMD Ryzen 3 or higher
Memory (RAM)	Minimum 4 GB (8 GB recommended)
Disk Space	Minimum 500 MB free space for JDK installation

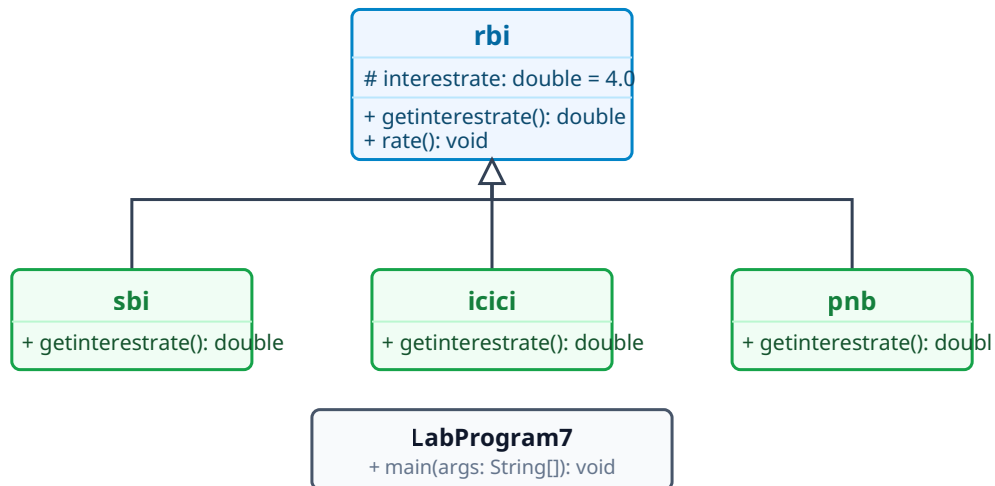
3. Algorithm

- Define Base Class (rbi):**
 - Declare a protected `double interestrate = 4.0`.
 - Define method `getinterestrate()` returning the interest rate (\$4.0%).
 - Define method `rate()` printing the current interest rate.
- Define Derived Classes:**
 - Create class `sbi` extending `rbi` and override `getinterestrate()` to return \$7.0%.
 - Create class `icici` extending `rbi` and override `getinterestrate()` to return \$8.3%.
 - Create class `pnb` extending `rbi` and override `getinterestrate()` to return \$6.6%.
- Execution Logic (LabProgram7 main method):**
 - Initialize `Scanner` object to read input.
 - Prompt user to enter bank name (e.g., "RBI", "SBI", "ICICI", "PNB").
 - Declare a reference variable `bank` of superclass type `rbi`.
 - Use a `switch` statement matching the uppercase bank name:
 - If "RBI" → Assign `bank = new rbi()`.
 - If "SBI" → Assign `bank = new sbi()`.
 - If "ICICI" → Assign `bank = new icici()`.
 - If "PNB" → Assign `bank = new pnb()`.
 - Default → Print "bank not found" and terminate program.

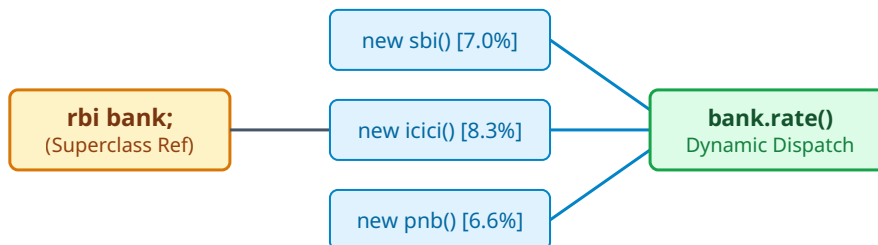
5. Call `bank.rate()`. Runtime polymorphism executes the overridden method corresponding to the instantiated object.
6. Close the scanner object and terminate execution.

4. UML Diagrams

4.1 Class Diagram



4.2 Polymorphic Dispatch Diagram



5. Java Source Code

```
import java.util.Scanner;

// Base Class representing Reserve Bank of India
class rbi {
    protected double interestrate = 4.0; // Standard interest rate guideline

    double getinterestrate() {
        return interestrate;
    }

    void rate() {
        System.out.println("rbi interest rate is" + getinterestrate() + "%");
    }
}
```

```

    }
}

// Derived Class SBI
class sbi extends rbi {
    @Override
    double getinterestrates() {
        return 7.0;
    }
}

// Derived Class ICICI
class icici extends rbi {
    @Override
    double getinterestrates() {
        return 8.3;
    }
}

// Derived Class PNB
class pnb extends rbi {
    @Override
    double getinterestrates() {
        return 6.6;
    }
}

// Main Program Class
public class LabProgram7 {
    public static void main(String[] args) {
        Scanner bi = new Scanner(System.in);
        System.out.println("enter the bank name to find the interest");
        String bankname = bi.nextLine();

        rbi bank; // Superclass reference variable

        switch (bankname.toUpperCase()) {
            case "RBI":
                bank = new rbi();
                break;
            case "SBI":
                bank = new sbi();
                break;
            case "ICICI":
                bank = new icici();
                break;
            case "PNB":
                bank = new pnb();
                break;
            default:
                System.out.println("bank not found");
                bi.close();
                return;
        }

        bank.rate(); // Dynamic Method Dispatch
        bi.close();
    }
}

```

6. Compilation & Execution Steps

Follow these command line steps to compile and run the Java program:

1. **Save Source File:** Create a file named `LabProgram7.java` and save the Java source code inside.
2. **Open Terminal/Command Prompt:** Navigate to the directory containing the file:

```
cd path/to/your/folder
```

3. **Compile Code:** Compile using the Java compiler (`javac`):

```
javac LabProgram7.java
```

Note: This produces class files: `rbi.class`, `sbi.class`, `icici.class`, `pnb.class`, and `LabProgram7.class`.

4. **Run Program:** Execute the compiled bytecode:

```
java LabProgram7
```

7. Output & Test Cases

Case 1: Querying SBI Interest Rate

```
enter the bank name to find the interest
sbi
rbi interest rate is7.0%
```

Case 2: Querying ICICI Interest Rate

```
enter the bank name to find the interest
ICICI
rbi interest rate is8.3%
```

Case 3: Querying PNB Interest Rate

```
enter the bank name to find the interest
pnb
rbi interest rate is6.6%
```

Case 4: Querying Base RBI Interest Rate

```
enter the bank name to find the interest
rbi
rbi interest rate is4.0%
```

Case 5: Invalid Bank Input

```
enter the bank name to find the interest
HDFC
```

bank not found